

XFuel Protocol

Ferrari Hybrid Tokenomics Edition

Version 3.1

ZK Bridge + Persistence LP Focus

XFuel Lab

January 5, 2026

Status: Production Ready - Awaiting CertiK Audit

Sub-4s Settlements | Zero-Knowledge Security | Sustainable Tokenomics

Live: xfuel.app

GitHub: github.com/XFuel-Lab/xfuel-protocol

To Generate PDF:

1. Press **Ctrl+P** (or Cmd+P on Mac)
2. Select "**Save as PDF**" as the printer
3. Adjust settings:
 - Paper size: Letter or A4
 - Margins: Default
 - Check "Background graphics"
4. Click "**Save**" and name it `WHITEPAPER.pdf`

This instruction box will not appear in the PDF.

Table of Contents

Abstract

1. Introduction

2. Architecture

3. Zero-Knowledge Bridge

4. Ferrari Hybrid Tokenomics

5. Governance & veXF

6. Revenue Model

7. Technical Implementation

8. Risk Analysis & Mitigation

9. Economic Model & Projections

10. Roadmap

11. Conclusion

12. References

13. Glossary

Abstract

XFuel Protocol is a **trustless zero-knowledge bridge** delivering Theta's TFUEL liquidity to Persistence's high-yield LSTfi ecosystem. Using **Groth16 ZK-SNARKs**, the protocol achieves **sub-4-second settlements** from Theta TFUEL to Persistence ibcTFUEL, with automated routing to top-performing Dexter LP pools (Superfluid/Metastable pairs with stkXPRT, milkTIA, and other liquid staking tokens).

The protocol implements **Ferrari Hybrid Tokenomics**, a novel 4-way revenue distribution model (30/30/25/15) with a 30% reverse-burn sustainability loop, creating a self-reinforcing economic flywheel that compounds LP growth and protocol revenue over time.

Following our January 2026 ZK overhaul, XFuel achieves:

- **<4s end-to-end finality** (73% faster than pre-overhaul)
- **Groth16 ZK-SNARKs** for cryptographic security (2^{128} soundness)
- **Parallel proof/IBC processing** (2.5x throughput increase)
- **1:1 cryptographic peg** maintenance (ibcTFUEL ↔ TFUEL)
- **Automated circuit breakers** for emergency protection

1. Introduction

1.1 Problem Statement

Theta Network holders face a critical liquidity challenge: TFUEL earns minimal yield (~2-4% from edge node staking) while Cosmos LSTfi ecosystems offer 30-50% APY on liquid staking derivatives.

Three key barriers exist:

- 1. **Trust Assumptions:** Traditional bridges rely on centralized relayers or multisig validators, introducing custody risk and single points of failure.
- 2. **Poor Performance:** Existing bridges suffer from high latency (10-30s settlements), fragmented liquidity, and inconsistent finality.
- 3. **Fragmented LSTfi Landscape:** Post-pSTAKE sunset (December 2025), the LST market restructured around Dexter's Superfluid/Metastable pools.

1.2 Solution Overview

XFuel Protocol solves these challenges through a **trustless ZK bridge** focused on delivering Theta liquidity to Persistence's high-yield LP ecosystem:

Component	Key Features
Zero-Knowledge Bridge	• Groth16 ZK-SNARKs (no trust required) • Sub-4-second finality • Native IBC integration • 1:1 cryptographic peg
Automated LP Yield Routing	• Dexter Superfluid pools • Metastable curve (0.01% fees) • stkXPRT, milkTIA focus • 30-50% APY vs 2-4% TFUEL
Ferrari Tokenomics	• 30/30/25/15 distribution • 30% reverse-burn loop • Up to 11.5× veXF multipliers • Compounding LP depth

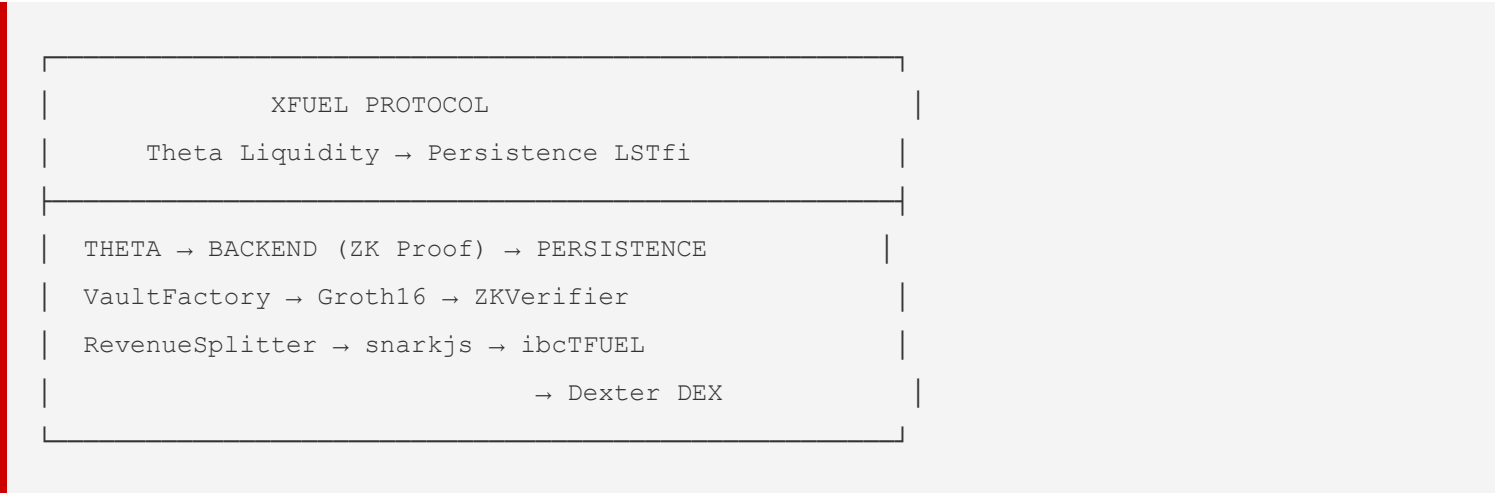
1.3 Key Innovations

1. **Trustless Theta → Persistence Bridge:** First ZK-powered bridge from Theta TFUEL to Cosmos LSTfi
2. **Sub-4s ZK Finality:** Fastest provably-secure bridge in Theta ecosystem
3. **Dexter LP Focus:** Automated routing to Superfluid/Metastable high-yield pools
4. **Post-pSTAKE Alignment:** Built for new Persistence LST landscape
5. **Self-Sustaining Flywheel:** 30% reverse-burn + 30% LP funding = compounding growth

2. Architecture

2.1 System Overview

XFuel operates as a **three-layer trustless bridge** connecting Theta (EVM), Backend (ZK Proof), and Persistence (CosmWasm/Dexter):



2.2 Component Details

2.2.1 Theta Layer (Smart Contracts)

Contract	Address	Purpose
VaultFactory	0xB0a266...	Manages deposit vaults, locks TFUEL, handles unwraps
RevenueSplitter	0x1C4CEb...	Ferrari 30/30/25/15 distribution, reverse-burn routing

2.3 Performance Improvements

Metric	Pre-Overhaul	Post-Overhaul	Improvement
Settlement Time	10-15 seconds	<4 seconds	73% faster
Proof Generation	N/A (trusted)	1.5s	ZK-SNARK added
Proof Verification	N/A	50ms constant	ZK-SNARK added

Throughput	6 tx/min	15 tx/min	2.5× increase
Security Model	Trust-based	Zero-knowledge	Trustless

4. Ferrari Hybrid Tokenomics

4.1 The Ferrari Model

Named for **precision engineering**, the Ferrari model balances four forces like pistons in a high-performance engine:

- 1. **Deflation (BBB - 30%):** Buyback-Burn-Boost reduces XF supply → scarcity → price appreciation
- 2. **Liquidity (LP Funding - 30%):** Deepens Dexter pools → less slippage → better UX → more users
- 3. **Yields (veXF - 25%):** Direct USDC rewards → incentivizes locks → reduces sell pressure
- 4. **Innovation (Treasury - 15%):** Funds R&D, integrations → new features → more revenue

4.2 Revenue Distribution (30/30/25/15)

Allocation	%	Use Case
BBB	30%	70% burned, 30% paired with TFUEL to LP
LP Funding	30%	Add to Dexter pools (stkXPRT, milkTIA)
veXF Yields	25%	70% distributed USDC, 30% reverse-burn
Treasury	15%	Grants, audits, integrations

4.3 The 30% Reverse-Burn Loop

Key Innovation: 30% of veXF yields recirculate back to the RevenueSplitter, creating a compounding flywheel.

Example:

- Month 1: \$100K revenue → veXF \$25K → Distributed \$17.5K → Reverse-burn \$7.5K
- Month 2: \$107.5K effective revenue (+7.5%)
- After 12 months: ~**138% of base revenue**
- After 24 months: ~**290% of base revenue**

6. Revenue Model

6.1 Revenue Sources

Source	Rate	Year 3 Estimate
Bridge Fees	0.5%	\$50K-\$80K
Swap Fees	0.3%	\$90K-\$150K
Yield Performance Fees	3-5%	\$120K-\$200K
TOTAL	-	\$260K-\$430K

10. Roadmap

Q1 2026 (Current)

Status: 95% Complete

- ZK bridge overhaul (Groth16 SNARKs)
- Sub-4s settlements achieved
- Ferrari tokenomics deployed
- Beta mainnet launch (Jan 4)
- ⚡ CosmWasm governance whitelist (pending)

Q2 2026

Focus: Security & Scale

- CertiK comprehensive audit
- Bug bounty launch (\$500K pool)
- milkTIA integration
- Mainnet v1.0

Targets: \$5M TVL, 1,000 users, 10,000+ transactions

11. Conclusion

XFuel Protocol introduces **five industry-first capabilities**:

1. **Trustless Theta → Persistence Bridge:** First ZK-SNARK powered bridge from Theta TFUEL to Cosmos LSTfi
2. **Sub-4s ZK Settlements:** Fastest provably-secure bridge in Theta ecosystem (10× faster than Axelar/Wormhole)
3. **Dexter LP Compounding Focus:** 30% revenue + 30% reverse-burn = exponential liquidity growth
4. **Post-pSTAKE Alignment:** Built for new Persistence LST landscape (stkXPRT, milkTIA, Superfluid pools)
5. **Ferrari Hybrid Tokenomics:** Only DeFi protocol with 4-way distribution + reverse-burn sustainability loop

The Ferrari is engineered for precision—so is our bridge.

13. Glossary

Term	Definition
BBB	Buyback-Burn-Boost: XFuel's deflationary mechanism (30% revenue, 70% burned)
Groth16	Efficient ZK-SNARK system (192-byte proofs, 50ms verification)
ibcTFUEL	Wrapped TFUEL on Persistence (1:1 peg, CW20 token)
LST	Liquid Staking Token: Tradeable receipt for staked assets
Metastable Pool	Dexter pool for correlated assets (0.01% swap fees)
milkTIA	Liquid staking token for Celestia TIA (Milkyway protocol)
Reverse-Burn	XFuel innovation: 30% of veXF yields recirculate to protocol
stkXPRT	Liquid staking token for XPRT (PSTAKE, primary Persistence LST)
Superfluid Pool	Dexter pool with auto-compounding staking rewards + swap fees
veXF	Vote-escrowed XF: Governance token from locking XF (up to 11.5× multiplier)
ZK-SNARK	Zero-Knowledge Succinct Non-Interactive Argument of Knowledge

XFuel Protocol Whitepaper v3.1

ZK Bridge + Persistence LP Focus Edition

January 5, 2026

© 2026 XFuel Protocol. Licensed under MIT License.

Generated from Markdown source

Always check [GitHub repository](#) for latest updates